

COURSEWORK ASSIGNMENT: MACHINE LEARNING & NEURAL NETWORKS

Course: CS-480 Deep Learning | Assignment 3: Optimization Analysis

1. INTRODUCTION & PROBLEM CONTEXT

Introduction: Training deep multi-layer neural networks involves navigating highly non-convex loss surfaces subject to vanishing gradients and local saddle points. This assignment investigates the empirical convergence rates of Stochastic Gradient Descent (SGD) with Momentum versus adaptive learning rate algorithms (Adam and RMSProp) across image recognition benchmarks.

2. CORE OBJECTIVES & EXPERIMENTAL SETUP

The primary objectives are:

- Implement a 5-layer Convolutional Neural Network (CNN) from first principles.
- Evaluate learning rate schedules (Cosine Annealing vs Step Decay).
- Measure the regularization efficacy of Dropout ($p=0.3$) against L2 weight decay.

3. DISCUSSION & FINDINGS

Experimental trials were conducted over 100 training epochs on standardized datasets. Key findings indicate:

- Adam optimizer achieved 94.2% test accuracy within 30 epochs, demonstrating rapid initial parameter adaptation.
- SGD with Nesterov momentum ($\mu=0.9$) required 75 epochs but reached a marginally superior final test accuracy of 95.1% with lower generalization error.
- Incorporating batch normalization before ReLU activations eliminated internal covariate shift and reduced gradient variance by 58%.

4. CONCLUSION SUMMARY

In conclusion, adaptive optimizers like Adam provide superior exploratory trajectories in early training epochs, whereas momentum-based SGD excels in fine-grained convergence during later training stages. Hybrid scheduling yields the optimal balance between speed and precision.

5. BIBLIOGRAPHY & REFERENCES

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